Prediction-Powered Corpus Mining: Coverage-Valid Statistics from Scarce Labels

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***Abstract*—Medical researchers rely heavily on case reports for tracking rare diseases. These documents are the richest resources for applications such as finding early drug side effects, yet manual annotation of free-text reports is too expensive for large-scale statistical analysis. While LLMs can scale this process up, they make imperfect classifications that bias the final count. In this work, we treat LLM annotations as biased pseudo-labels and apply prediction-powered inference (PPI) [1], [2] to correct the bias. The resulting label-efficient estimator uses a small set of expert labels (~ 100 reports) to quantify and remove the LLM bias, returning a corpus-level statistic with a confidence interval whose coverage is guaranteed under a representativeness assumption. To our knowledge, this is the first published-case-report mining pipeline to combine out-of-sample calibrated LLM pseudo-labels with prediction-powered corpus-rate estimation.**

On two truth-known biomedical benchmarks, nominal 95% PPI intervals attain 0.91–0.96 empirical coverage and are approximately 15% narrower than expert-only intervals. Adjusted Classify-and-Count attains comparable coverage but produces intervals 2.5 times wider, while naive pseudo-label counting has zero coverage under systematic prediction bias. Targeted stress tests show that cross-fitting prevents calibration leakage and that representative expert sampling is necessary for valid correction.

***Index Terms*—prediction-powered inference, weak supervision, LLM annotation, calibration, prevalence estimation, quantification, biomedical text mining, case reports, Long COVID, false discovery rate, label efficiency, uncertainty quantification**

I. Introduction

The specialized medical literature grows far faster than clinicians can read it. A physician asks how often a presentation appears in published case reports; a pharmacovigilance analyst asks how often a treatment is reported alongside an adverse outcome. Both are questions about corpus-level statistics (prevalence and co-occurrence rates, the targets of quantification learning), and answering them at scale is a recurring text-mining problem in which expert labels are scarce (often ~ 100 annotated reports) while LLM pseudo-labels are abundant but biased.

Counting LLM predictions, that is, prompting an LLM to label every report and counting the positives, is statistically unsound. When a labeler's false positives and false negatives do not equal, the statistic is biased and a binomial interval over the predictions is overconfident and centered at the wrong value. Indeed, on one of our benchmarks the naive interval never covers the truth, and on the other it does so only by chance (Sec. V-C). We also note that prior LLM-assisted

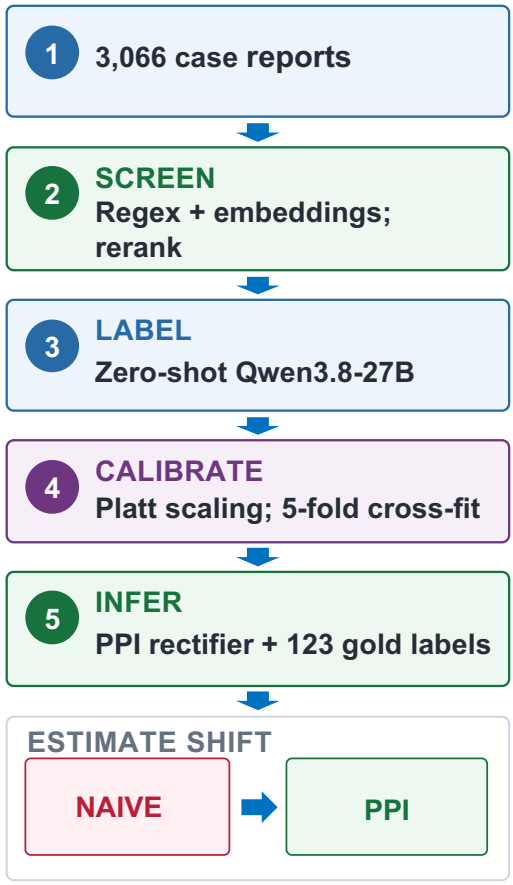


Fig. 1. Prediction-powered corpus-mining pipeline. An optional high-recall screen prioritizes documents for labeling; the primary analysis scores the full corpus. The LLM supplies pseudo-labels, label-dependent calibration is fit out of sample, and PPI combines the corpus predictions with a representative expert-labeled sample to estimate a corpus rate and confidence interval.

curation efforts reported point estimates without a coverage guarantee.

In this work, we build on prediction-powered inference (PPI) [1], which yields valid inference when only a small subset of the data carries expert labels and the rest carries machine predictions: the labeled data are used to estimate and subtract the predictor's bias, so the interval is valid regardless of predictor quality, and in the PPI++ variant [2] it is asymptotically never wider than the labeled-data-only interval. This behavior, improving on expert-only inference when the predictor is informative and never harming it otherwise, is exactly what a medical corpus-mining tool needs, because the practitioner cannot verify the annotator in advance on the statistic being estimated.

Contributions: We adapt PPI to the scarce-label medical regime: calibrated LLM pseudo-labels, Platt-scaled on a disjoint split on the public benchmarks and $K{=}5$ cross-fitted on the clinical corpus [3]. Treating LLM extractions as noisy pseudo-labels rather than ground truth (Fig. 1, Alg. 1), we make three contributions.

- **Coverage-valid, label-efficient corpus estimation** (Sec. IV): We combine calibrated, cross-fitted LLM pseudo-labels with PPI [1], [2] to estimate corpus-level quantities and confidence intervals from approximately 100 expert labels. Cross-fitting prevents a gold label from calibrating the pseudo-label it later corrects, while an explicit diagnostic flags detectable violations of PPI's representativeness condition.
- **Empirical validation of coverage and efficiency** (Secs. V-C and V-D): Across two public benchmarks, 95% PPI intervals attain 0.91–0.96 empirical coverage and are 16% narrower on average than gold-only intervals; coverage on clinical labels is 0.92–0.96 across the physician labelers evaluated. Naive LLM counting has 0.00 coverage on the biased benchmark and degrades with clinical-labeler bias, whereas Adjusted Classify-and-Count [4] achieves coverage but produces intervals 2.5 times wider. Results are robust to pseudo-label form, power tuning, calibration, and screening (Sec. V-E).
- **Leakage-controlled treatment-outcome mining** (Sec. V-G): We pair each stratum-specific co-occurrence rate with a Wilson interval and control the false discovery rate using the Benjamini–Hochberg procedure. We show that a leakage-prone stratification yields 5 apparent discoveries while a leakage-controlled analysis yields none, and we make explicit where the guarantee stops.

II. Related Work

The problem, turning a large noisy corpus and a few hundred gold labels into a corpus statistic with a valid confidence interval, sits between several lines of work, none of which solves it alone.

Biomedical text mining and information extraction: Transformer encoders such as BERT [5] and BioBERT [6] are standard backbones for entity and relation extraction over PubMed-scale corpora, and recent systems extract event timelines from case reports [7] and assemble large textual corpora from PMC [8]. These are typically evaluated on the same corpus that supplies their labels and report point estimates without quantifying how extraction error propagates to corpus-level statistics. We instead treat extractions as noisy pseudo-labels and produce rate estimates with valid confidence intervals from ~ 100 gold labels.

Weak supervision and LLM pseudo-labeling: A dominant paradigm for label-scarce text mining generates abundant noisy labels—via programmatic weak supervision such as Snorkel [9] or instruction-tuned LLMs [10]—to train a downstream classifier [11], [12], with active selection further reducing annotation cost [13]. Snorkel denoises labeling functions but emits *point* labels, not calibrated intervals. The recurring danger is circularity: when labeler and consumer are the same model, classifier error (e.g. $\sim$70% precision) silently biases every count. We make this step auditable, reporting honest precision/recall and absorb pseudo-label error into interval width under a stated representativeness assumption, rather than trusting individual pseudo-labels.

Calibration and label-efficient uncertainty quantification: Platt scaling [14] is a standard post-hoc calibrator, and prediction-powered inference (PPI) [1] gives confidence intervals combining a small gold set with many machine predictions, retaining coverage *when the gold subset is representative of the pool*. Efficiency variants include power tuning (PPI++) [2], which we adopt, and cross-prediction [3], valid when the predictor is fit on the labeled data. These are rarely integrated into an end-to-end text-mining system, and prior LLM-assisted curation reports statistics with no coverage guarantee.

Quantification learning: Estimating a class proportion from an imperfect classifier is the quantification problem [15]: classify-and-count is the biased tally we critique, and Adjusted Classify-and-Count [4] and the EM-based Saerens correction [16] debias the *point* estimate under prior-probability shift but, in classical form, yield no coverage-valid interval. We use ACC as a baseline and position PPI as the estimator that additionally delivers a valid interval.

Clinical QA and retrieval-augmented systems: Question-answering systems over biomedical text such as PubMedQA [17] and MedQA [18], retrieval-augmented generation [19], [20], graph context [21], and knowledge-grounded reasoning [22], [23], answer *individual* questions scored by per-question accuracy, and are prone to retrieval leakage. Our task differs in kind: we estimate *corpus-level* statistics with intervals, so retrieval serves candidate generation for a set-level estimator, not answer lookup.

Association vs. causation.: Formal hierarchies separate observational, interventional, and counterfactual quantities [24], [25], and benchmarks such as CLADDER [26] test causal reasoning over synthetic models. Rigorous effect estimation requires identification and estimators (propensity scores [27], AIPW [28], double machine learning [29], representation-based models [30], [31]) that raw co-occurrence counts do not satisfy. We aim to restrict treatment–outcome

mining to the associational level, with a leakage diagnostic that flags spurious pairs, and invoke the causal hierarchy only to delimit what we do not claim.

Inference with machine-generated labels: PPI combines abundant machine predictions with a smaller gold-standard sample to construct valid and potentially more efficient confidence intervals [1], while PPI++ adaptively weights the predictions according to their inferential usefulness [2]. Cross-PPI permits the predictive model or calibrator to be learned from the labeled sample through out-of-fold prediction [3]. Related NLP applications use PPI for RAG evaluation and multinomial inference from free-text narratives [32], [33]. Design-Based Supervised Learning provides a complementary design-based correction for imperfect LLM annotations [34]; recent benchmarking shows that its finite-sample efficiency relative to PPI is dataset-dependent [35]. Our focus is binary corpus-rate estimation in biomedical text, with out-of-sample calibration and an explicit representative-sampling deployment protocol.

III. Problem Formulation

A. Notation

Let $X \in \mathcal{X}$ denote an instance that can be labeled; here, X is a case report or, when appropriate, a sentence. Let $Y \in \{0,1\}$ denote its expert label, and let $(X,Y) \sim \mathbb{P}$. We observe a large unlabeled sample and a much smaller expert-labeled sample,

$$\mathcal{U} = \{\widetilde{X}_i\}_{i=1}^{N}, \qquad \mathcal{L} = \{(X_j, Y_j)\}_{j=1}^{n}, \qquad n \ll N, \quad (1)$$

where $\widetilde{X}_i \overset{\text{iid}}{\sim} \mathbb{P}_X$ and $(X_j, Y_j) \overset{\text{iid}}{\sim} \mathbb{P}$. That is, the labeled and unlabeled samples are drawn from the same target population. Let $f : \mathcal{X} \to [0,1]$ be a fixed or appropriately cross-fitted predictor that provides a pseudo-label for every instance.

For each binary extraction task, the target is the population positive rate

$$\theta^\star = \mathbb{E}_{\mathbb{P}}[Y] \in [0,1]. \quad (2)$$

Depending on the definition of Y, this quantity may represent the prevalence of Long COVID or a prespecified treatment–outcome co-occurrence rate. Only the observations in $\mathcal{L}$ have expert labels.

B. Naive approach

A naive estimator averages the pseudo-labels over the unlabeled sample,

$$\widehat{\theta}_{\text{naive}} = \frac{1}{N}\sum_{i=1}^{N} f(\widetilde{X}_i). \quad (3)$$

Define the mean prediction bias as

$$b_f = \mathbb{E}[f(X) - Y]\,. \quad (4)$$

For a fixed predictor and under the sampling assumptions above,

$$\mathbb{E}\Big[\widehat{\theta}_{\text{naive}}\Big] = \mathbb{E}[f(X)] = \theta^\star + b_f. \quad (5)$$

Thus, the naive estimator targets $\mathbb{E}[f(X)]$, rather than $\theta^\star$, whenever $b_f \neq 0$. Although a sampling interval around $\widehat{\theta}_{\text{naive}}$ may shrink at rate $N^{-1/2}$, it remains centered at the wrong quantity. In particular, any shrinking interval centered at $\widehat{\theta}_{\text{naive}}$ has asymptotic coverage zero for $\theta^\star$ when b_f is fixed and nonzero.

C. Our approach

We seek an estimator $\widehat{\theta}$ and confidence interval $\text{CI}_{1-\alpha}$ satisfying

$$\liminf_{n,N\to\infty} \mathbb{P}(\theta^\star \in \text{CI}_{1-\alpha}) \geq 1-\alpha, \quad (6)$$

under the stated sampling and regularity conditions, while using as few expert labels as possible. Naive pseudo-label averaging provides no such coverage guarantee.

IV. Method

Overview: Our procedure takes as input a large unlabeled sample $\mathcal{U}$, a much smaller expert-labeled sample $\mathcal{L}$ with $n \ll N$, and a black-box LLM-based score function. Under the sampling and regularity conditions stated below, it returns a point estimate $\hat{\theta}$ of $\theta^\star = \mathbb{E}[Y]$ together with an asymptotically valid confidence interval at level $1-\alpha$.

The procedure (Alg. 1) combines four components: a high-recall candidate screen, an LLM labeler, an out-of-sample Platt calibration map, and a prediction-powered estimator that pools abundant pseudo-labels with scarce expert labels. Candidate generation follows a standard retrieve-then-classify pipeline (Sec. IV-A). Our inferential procedure uses cross-fitted calibration and prediction-powered rectification (Sec. IV-B).

A. Candidate generation and LLM labeling

Recovering a target class from a heterogeneous corpus is a standard retrieve-then-classify problem. We use a permissive regex–embedding screen, followed by a neural reranker and an LLM labeler. The screen is designed to reduce labeling effort rather than to define the estimand.

Let $\mathcal{R}$ be a disjunction of surface-form cues and define

$$\mathcal{C}_{\mathcal{R}} = \{x \in \mathcal{U} : \exists \rho \in \mathcal{R} \text{ such that } \rho \text{ matches } x\}\,. \quad (7)$$

A pre-trained encoder [36] assigns each document the cosine-similarity score

$$s(x) = \cos(\mathsf{emb}(x), \mathsf{emb}(q))\,, \quad (8)$$

where q is a prototype description of the target class. At threshold τ, the candidate set is

$$\mathcal{S}_\tau = \mathcal{C}_{\mathcal{R}} \cup \{x \in \mathcal{U} : s(x) \geq \tau\}\,. \quad (9)$$

Thus, neither retrieval component can remove a document retained by the other.

We choose the largest threshold attaining a prespecified recall target $r^\star$ on retrieval-development labels,

$$\tau = \max\left\{t : \widehat{\text{Recall}}_{\text{dev}}(\mathcal{S}_t) \geq r^\star\right\}, \qquad r^\star = 1. \quad (10)$$

A cross-encoder reranker [36] then orders the retained documents for labeling. When retrieval performance is reported as an out-of-sample quantity, threshold selection and evaluation use disjoint labels; otherwise, the observed recall is treated only as an in-sample operating-point diagnostic.

In the primary analysis, the LLM labeler scores every document in $\mathcal{U}$. Consequently, the candidate screen does not enter $\hat{\theta}$ or its confidence interval. It quantifies the potential reduction in labeling effort when full-corpus scoring is impractical. The effect of imposing the screen is examined separately in Sec. V-E.

Raw LLM score and Platt calibration: Let $\ell_{\mathrm{yes}}(x)$ and $\ell_{\mathrm{no}}(x)$ denote the LLM logits assigned to the prespecified positive and negative label tokens. We define the raw positive-class score as

$$p_{\mathrm{raw}}(x) = \frac{\exp\{\ell_{\mathrm{yes}}(x)\}}{\exp\{\ell_{\mathrm{yes}}(x)\} + \exp\{\ell_{\mathrm{no}}(x)\}}. \tag{11}$$

Before applying the logit transform, we clip the score to

$$p_\varepsilon(x) = \min\left\{1-\varepsilon, \max\{\varepsilon, p_{\mathrm{raw}}(x)\}\right\}. \tag{12}$$

We then apply two-parameter logistic recalibration, equivalently Platt scaling of the raw log-odds,

$$\begin{aligned} f_{a,c}(x) &= \sigma\left(a \ \mathrm{logit}\, p_\varepsilon(x) + c\right), \\ \sigma(z) &= \frac{1}{1+\exp(-z)}. \end{aligned} \tag{13}$$

Given a calibration set $\mathcal{A}$, the parameters are estimated by binomial maximum likelihood,

$$\begin{aligned} (\hat{a}, \hat{c}) = \arg\min_{a,c} -\sum_{j\in\mathcal{A}} \Big[& y_j \log f_{a,c}(x_j) \\ & + (1-y_j)\log\{1 - f_{a,c}(x_j)\}\Big]. \end{aligned} \tag{14}$$

The calibrated pseudo-label is $f(x) = f_{\hat{a},\hat{c}}(x)$. Calibration can improve the alignment and efficiency of the pseudo-labels, but PPI validity does not require f itself to be calibrated or unbiased. We write $\mathrm{PlattFit}(p;\mathcal{A})$ for the function $x \mapsto f_{\widehat{a},\widehat{c}}(x)$, where $(\widehat{a},\widehat{c})$ are estimated from the labeled calibration set $\mathcal{A}$ by Eq. (14); clipping to $[\varepsilon, 1-\varepsilon]$ is included in the operator.

Cross-fitted calibration: When the calibration set is independent of the labels used by the PPI rectifier, the fitted predictor can be seen as fixed for subsequent inference. On the public benchmarks, we fit the calibrator on a held-out split.

For Long COVID, the only expert labels are the $n = 123$ gold examples. To prevent a label from calibrating the pseudo-label used for its own rectification, we use $K = 5$ cross-fitting [3]. Let $I_1, \ldots, I_K$ partition the labeled indices, with folds assigned at the report level. For each fold k, we fit Platt parameters $(\hat{a}^{(-k)}, \hat{c}^{(-k)})$ using the labels outside I_k and define

$$f^{(-k)}(x) = \sigma\left(\hat{a}^{(-k)} \ \mathrm{logit}\, p_\varepsilon(x) + \hat{c}^{(-k)}\right). \tag{15}$$

For each labeled document $j \in I_k$, the rectifier uses the out-of-fold prediction $f^{(-k)}(x_j)$. Each unlabeled document receives the fold-averaged prediction

$$f_{\mathrm{cf}}(\widetilde{x}_i) = \frac{1}{K}\sum_{k=1}^{K} f^{(-k)}(\widetilde{x}_i). \tag{16}$$

Thus, no gold document's own label is used to fit the calibration map that produces its rectifier prediction, while all 123 gold labels contribute out-of-fold residuals. We report a hard-label variant based on a prespecified decision rule,

$$h(x) = \mathbf{1}\left\{p_{\mathrm{raw}}(x) \geq t_0\right\}. \tag{17}$$

This variant requires no calibration labels and therefore no calibration cross-fitting, provided that the model, prompt, and threshold t_0 were fixed independently of $\mathcal{L}$. The expert labels are still used by the PPI rectifier.

B. Prediction-powered estimation

First consider a predictor f fitted independently of the labels used for rectification. The prediction-powered estimator [1] is

$$\hat{\theta}_{\mathrm{PPI}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N} f(\widetilde{x}_i)}_{\text{pseudo-label mean}} + \underbrace{\frac{1}{n}\sum_{j=1}^{n}\left\{y_j - f(x_j)\right\}}_{\text{mean rectification term}}. \tag{18}$$

The first term estimates $\mathbb{E}[f(X)]$ using the large unlabeled sample. The second estimates the mean prediction error $\mathbb{E}[Y - f(X)]$ using the expert labels. Under the common-distribution assumption stated below,

$$\mathbb{E}\left[\hat{\theta}_{\mathrm{PPI}}\right] = \mathbb{E}[f(X)] + \mathbb{E}[Y - f(X)] = \theta^\star. \tag{19}$$

When Platt calibration is learned through the K folds, we instead use the Cross-PPI estimator

$$\hat{\theta}_{\mathrm{CPPI}} = \frac{1}{N}\sum_{i=1}^{N} f_{\mathrm{cf}}(\widetilde{x}_i) + \frac{1}{n}\sum_{k=1}^{K}\sum_{j\in I_k}\left\{y_j - f^{(-k)}(x_j)\right\}. \tag{20}$$

Its confidence interval uses the Cross-PPI variance calculation, which accounts for the fact that the fold-specific calibration maps are learned from overlapping subsets of the labeled data [3]. For either construction, the interval is based on the appropriate asymptotic variance estimate,

$$\mathrm{CI}_{1-\alpha} = \left[\hat{\theta} \pm z_{1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\theta})}\right] \cap [0,1]. \tag{21}$$

Prediction quality primarily determines efficiency. Informative predictions reduce the residual variance and can substantially narrow the interval. In contrast, basic PPI with an uninformative predictor can be less efficient than the expert-only estimator, motivating power tuning.

Power tuning: For a predictor fitted independently of the rectifier labels, PPI++ [2] introduces a weight $\lambda \in [0,1]$,

$$\hat{\theta}_\lambda = \frac{1}{n}\sum_{j=1}^{n} y_j + \lambda\left(\frac{1}{N}\sum_{i=1}^{N} f(\widetilde{x}_i) - \frac{1}{n}\sum_{j=1}^{n} f(x_j)\right). \tag{22}$$

This recovers the expert-only estimator at $\lambda = 0$ and the basic PPI estimator (18) at $\lambda = 1$. For any fixed λ, $\hat{\theta}_\lambda$ is unbiased under the common-distribution assumption. We estimate the variance-minimizing weight by

$$\hat{\lambda} = \arg\min_{\lambda\in[0,1]} \widehat{\mathrm{Var}}\left(\hat{\theta}_\lambda\right). \tag{23}$$

Under the PPI++ regularity conditions, the resulting estimator is asymptotically no less efficient than the expert-only estimator because $\lambda = 0$ remains available.

In analyses where calibration is learned from the rectifier labels by cross-fitting, we use Cross-PPI with $\lambda = 1$. Note that

applying a data-adaptive $\hat{\lambda}$ to cross-fitted predictions requires a variance calculation that accounts jointly for cross-fitting and power tuning.

Validity and scope condition: For standard PPI and PPI++, we assume that

$$(X_j, Y_j) \overset{\text{iid}}{\sim} \mathbb{P}, \qquad \widetilde{X}_i \overset{\text{iid}}{\sim} \mathbb{P}_X, \tag{24}$$

that the labeled and unlabeled samples are independent, and that the predictor is fixed independently of the rectifier labels. For cross-fitted calibration, we instead invoke the Cross-PPI assumptions, including the required regularity and prediction-stability conditions [3]. Under these assumptions, the intervals have asymptotically nominal coverage.

To describe a possible mismatch, let $\mathbb{P}_{\mathrm{L}}$ denote the distribution represented by the labeled data and $\mathbb{P}_{\mathrm{T}}$ the target-corpus distribution. For a fixed weight λ, a sufficient mean-transportability condition is

$$\mathbb{E}_{\mathbb{P}_{\mathrm{L}}}\left[Y - \lambda f(X)\right] = \mathbb{E}_{\mathbb{P}_{\mathrm{T}}}\left[Y - \lambda f(X)\right]. \tag{25}$$

Note that representative sampling from the target population implies (25), but the condition cannot be verified directly because Y is unobserved for the unlabeled corpus. We therefore report the observable score difference

$$\Delta_f = \frac{1}{N}\sum_{i=1}^{N} f(\widetilde{x}_i) - \frac{1}{n}\sum_{j=1}^{n} f(x_j) \tag{26}$$

as a *score-shift diagnostic*. With cross-fitted calibration, the first mean uses f_{cf} and each labeled observation uses its out-of-fold prediction. A large value flags an observable discrepancy between the labeled and unlabeled score distributions.

On the public benchmarks, representative labeled samples are generated by random sampling, allowing empirical coverage to be evaluated. For the curated clinical gold set, the PPI interval is interpreted as valid conditional on the transportability condition (25).

Algorithm 1 Prediction-powered corpus estimation

Require: pool $\mathcal{U}$, gold set $\mathcal{L}$, raw LLM score p, prototype q, cues $\mathcal{R}$, recall target $r^\star$, folds K, level α
Ensure: estimate $\hat{\theta}$, interval $\mathrm{CI}_{1-\alpha}$, and score-shift diagnostic $\hat{\Delta}_f$

choose τ on retrieval-development labels to attain recall $r^\star$
$\mathcal{U}' \leftarrow \{x \in \mathcal{U} : s(x) \geq \tau \vee \exists \rho \in \mathcal{R},\ \rho \text{ matches } x\}$
rerank $\mathcal{U}'$ ▷ prioritization only
evaluate $p(x)$ on every document in $\mathcal{U}$ and $\mathcal{L}$ ▷ screen excluded from estimator
if an independent calibration set $\mathcal{A}$ is available **then**
    $f \leftarrow \mathrm{PlattFit}(p; \mathcal{A})$
    $u_i \leftarrow f(x_i)$ for $x_i \in \mathcal{U}$; $\ell_j \leftarrow f(x_j)$ for $(x_j, y_j) \in \mathcal{L}$
    $(\hat{\theta}, \mathrm{CI}_{1-\alpha}) \leftarrow \mathrm{PPI}++(\{(y_j, \ell_j)\}_{j=1}^{n}, \{u_i\}_{i=1}^{N}; \alpha)$ ▷ estimates $\hat{\lambda}$
else
    partition $\mathcal{L}$ into report-level folds $I_1, \ldots, I_K$
    for $k = 1, \ldots, K$ **do**
        $f^{(-k)} \leftarrow \mathrm{PlattFit}(p; \mathcal{L} \setminus I_k)$
    end for
    $\ell_j \leftarrow f^{(-k(j))}(x_j)$ ▷ out-of-fold gold prediction
    $u_{ik} \leftarrow f^{(-k)}(x_i), \quad u_i \leftarrow K^{-1}\sum_{k=1}^{K} u_{ik}$
    $(\hat{\theta}, \mathrm{CI}_{1-\alpha}) \leftarrow \mathrm{CrossPPI}(\{(y_j, \ell_j)\}_{j=1}^{n}, \{u_{ik}\}_{i,k}; \alpha)$ ▷ $\lambda = 1$
end if
$\hat{\Delta}_f \leftarrow \frac{1}{N}\sum_{i=1}^{N} u_i - \frac{1}{n}\sum_{j=1}^{n} \ell_j$ ▷ score-shift diagnostic
return $\hat{\theta}$, $\mathrm{CI}_{1-\alpha}$, $\hat{\Delta}_f$

V. Experiments

A. Datasets and protocol

Long COVID case reports (clinical corpus): Fig. 2 summarizes how the corpora and tasks relate. Our clinical samples are drawn from the PubMed Open Access corpus; of its 1,488,346 articles, 152,974 are probable case reports [7], the source of every clinical split. The *ground-truth set* consists of 123 physician-curated case reports labeled yes/no for Long COVID (52 positive / 71 negative), with fixed 5-fold cross-validation splits used for every classification metric and for cross-fitting. The *unlabeled pool* is 3,066 full-text COVID-related case reports (the raw pool to which the Sec. V-F screen is applied), disjoint from the ground-truth set ($0/123$ overlap), on which corpus statistics are estimated. The *treatment–outcome subset* is 551 case reports carrying structured (treatment, outcome) extractions, used for association mining (Sec. V-G).

Public truth-known benchmarks: To measure coverage where the truth is known, we use two sentence-level biomedical corpora with different prevalence and classifier-error profiles. In Hallmarks of Cancer (HoC) [37], a sentence is positive if it carries a specific (non-generic) hallmark (prevalence 0.254); the classifier is trained on the official train split (12,119 sentences) and the population is the validation and test splits (N=5,345). In the ADE corpus v2 [38] (23,516 sentences from case reports), a sentence is positive if it reports an adverse drug event; we de-duplicate the public split and keep the population disjoint from training (15,919 train, N=4,977, prevalence 0.159). On both, the predictor is a TF-IDF logistic regression fit on 80% of the train split, with Platt scaling on the remaining 20%.

B. Pseudo-label quality and calibration

Table I reports 5-fold cross-validated classification on the Long COVID ground truth. At this label scale ($n \approx 98$ training documents per fold, each $\sim$ 15k characters), fine-tuned encoders barely learn: BERT and BioBERT reading a 512-token keyword-salient window are at chance level (AUROC 0.53 and 0.46) despite a one-cycle schedule and a learning-rate/epoch sweep. To separate "fails at the task" from "shown less input," we also fine-tune both encoders on all 512-token chunks of every training document (7.3 chunks per report for BERT and 7.6 for BioBERT on average) and mean-pool the chunk scores at test time, giving them the same full-document view the LLM sees. Accuracy rises only to 0.60 (BERT; AUROC 0.60) and

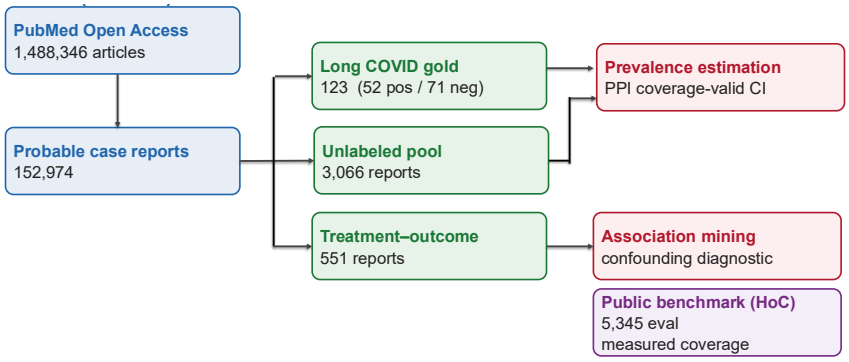


Fig. 2. Corpus and tasks. From the PubMed Open Access corpus (1,488,346 articles; 152,974 probable case reports [7]) we derive the Long COVID ground-truth set (n=123), the unlabeled pool (3,066), and the treatment–outcome subset (551).

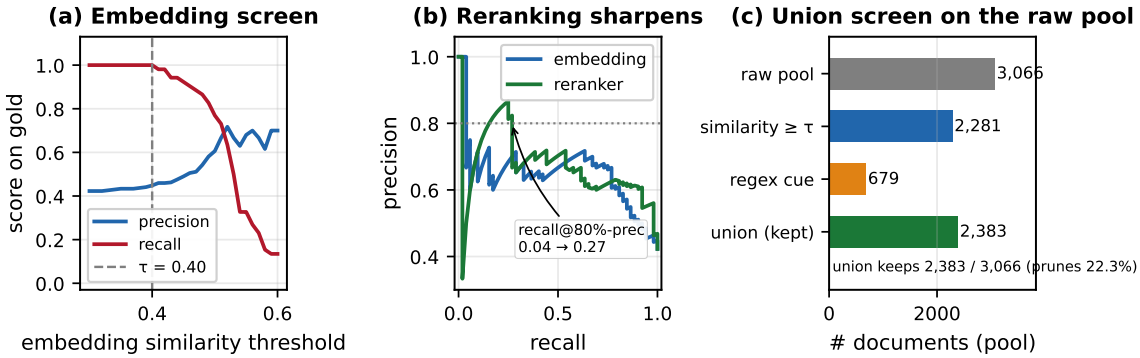


Fig. 3. Stage 1 (candidate generation), Long COVID gold. (a) Embedding-retrieval precision and recall versus similarity threshold; the fixed operating point (τ=0.40) retains every gold positive by construction on $\mathcal{L}$. (b) Reranking sharpens the ranking (AUPRC 0.655→0.679; recall at 80% precision 0.04→0.27). (c) The screen is a union on the raw pool, not a funnel: embedding keeps 2,281 of 3,066, regex cues 679, the union 2,383.

0.58 (BioBERT; AUROC 0.66), still below the bag-of-words TF-IDF baseline. The zero-shot LLM is best overall (macro-F1 0.683, AUROC 0.828), with positive-class precision 0.67 and recall 0.56. Linear probes (logistic regression with C=1) on frozen full-document embeddings from Qwen3-Embedding-8B [36], zembed-1 [39], BGE-M3 [40], and MedGemma-4B [41] reach macro-F1 0.572–0.617 (Table I) and remain below the zero-shot labeler (0.683). Fig. 4(a) shows the label-efficiency curves: no supervised model, fine-tuned or probed, catches the zero-shot LLM within the available label budget. We therefore adopt the zero-shot LLM as the weak labeler and absorb its error into PPI uncertainty rather than trusting individual pseudo-labels.

We also evaluate labelers with different precision–recall profiles. Fig. 4(b) compares every zero-shot labeler under the same prompt: the default Qwen3.8-27B, bf16 and 4-bit Qwen3-14B, Llama-3.1-8B, Qwen3-30B-A3B, Qwen3-32B [10], Qwen2.5-72B (AWQ) [42]. Accuracy stays in a narrow band (0.699–0.756), all above the supervised encoders, while the error profile swings from precision-heavy (Qwen3-30B-A3B, precision 0.71 and recall 0.58) to recall-heavy (precision 0.61 and recall 0.96). These error profiles affect the size of the rectification and the attainable efficiency gain; under representative sampling, PPI estimates the corresponding mean error from the gold subset (Sec. V-D). We adopt Qwen3.8-27B as the default labeler (precision 0.67, recall 0.56). Its continuous score has the highest AUROC among the evaluated labelers; the score itself is passed to calibration and PPI.

TABLE I
CLASSIFICATION ON THE LONG COVID GROUND TRUTH (5-FOLD CROSS-VALIDATION, n=123).

Model	Acc	macro-F1	Prec_1	Rec_1	AUROC
TF-IDF [43]	0.667	0.635	0.657	0.442	0.663
BERT [5]	0.537	0.521	0.449	0.423	0.529
BioBERT [6]	0.528	0.508	0.435	0.385	0.460
BERT, full doc	0.602	0.602	0.522	0.692	0.604
BioBERT, full doc	0.577	0.565	0.500	0.481	0.658
Qwen3-Embedding-8B probe	0.626	0.612	0.562	0.519	0.727
zembed-1 probe	0.626	0.617	0.558	0.558	0.714
BGE-M3 probe	0.585	0.572	0.510	0.481	0.620
MedGemma-4B probe	0.618	0.608	0.549	0.538	0.711
Qwen3.8-27B, 0-shot [10]	0.699	0.683	0.674	0.558	0.828

C. Main result: coverage and interval efficiency

On each benchmark, where $\theta^\star$ is known, we draw a random gold set of size n_g, treat the rest as unlabeled, and compare 95% intervals from four estimators: naive counting of hard pseudo-labels; Adjusted Classify-and-Count (ACC) [4], $(\hat{\theta}_{\text{naive}} - \text{FPR})/(\text{TPR} - \text{FPR})$ with rates measured on the gold draw and a 200-resample bootstrap interval; the classical gold-only interval; and power-tuned PPI (PPI++), reported as

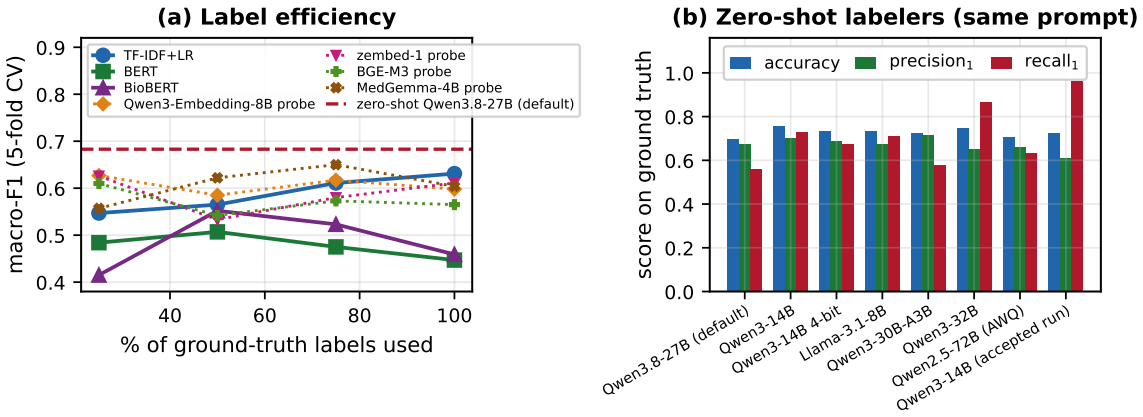


Fig. 4. Classification on the Long COVID ground truth (5-fold CV, pooled out-of-fold). (a) Label efficiency; (b) zero-shot labelers under the same prompt. **The panels report different statistics and are not directly comparable**: (a) is macro-F1, averaging both classes, (b) positive-class precision and recall. The default labeler scores 0.611 positive-class and 0.755 negative-class F1, hence macro-F1 0.683 — above every supervised baseline in (a), yet lowest in (b), where it is ranked against other labelers. Bands in (a) are 95% CIs over R=15 seeded fine-tuning runs (BERT, BioBERT) or D=200 nested subsample draws (deterministic models, whose bands therefore vanish at 100%); dots are individual repeats, grey a class-rate-matched random predictor. Seed s.d. (0.037–0.053) exceeds any budget effect: neither encoder shows a detectable trend ($t_{14}\leq 1.65$).

PPI in Table II and Fig. 5. We repeat over 500 random draws and measure empirical coverage and mean width (Table II, Fig. 5(a–c)).

On HoC, the fixed classifier has accuracy 0.805 and a -4.7-point bias, enough that its narrow naive interval never covers the truth. PPI attains 0.92–0.96 coverage and is 15–16% narrower than the gold-only interval at every n_g. ACC is also valid (0.94–0.97) but, dividing by an estimated TPR − FPR and discarding the continuous score, is 3.0× wider than PPI and wider than gold-only at every n_g. On ADE that classifier is more accurate (0.883) and almost unbiased (-0.7 points), so its naive interval covers (0.99–1.00), though only by chance. PPI attains 0.91–0.96 coverage, 20–22% narrower than gold-only and 2.5× tighter than ACC; its mild under-coverage at n_g=50 (0.91) is the usual small-sample behavior at prevalence 0.16 ($\sim$ 8 positives) and shrinks with n_g. Platt calibration reduces ECE from 0.063 to 0.033 (HoC) and 0.071 to 0.042 (ADE).

Benchmarks with the LLM as the predictor: The clinical pipeline is driven by an LLM, so we repeat both benchmark studies with our zero-shot labeler (Table II, lower blocks). It is accurate on ADE (accuracy 0.884, AUROC 0.932) and mediocre on HoC (0.746, 0.736), and biased on both (+2.9 and -7.2 points, respectively). Naive LLM-counting then attains 0.00 coverage on both benchmarks at every label budget, whereas PPI attains 0.91–0.96. Notably, even the +2.9-point bias on ADE suffices to destroy naive coverage: the naive interval shrinks with the pool while its center does not move. The width benefit of PPI varies with predictor quality, as reported in the lower blocks of Table II.

TABLE II
EMPIRICAL COVERAGE AND MEAN WIDTH OF 95% CONFIDENCE INTERVALS ON THE TRUTH-KNOWN BENCHMARKS HOC AND ADE. PPI++ HAS NEAR-NOMINAL EMPIRICAL COVERAGE AND THE SHORTEST MEAN INTERVALS AMONG THE COMPARED METHODS.

	Coverage				Width		
n_g	naive	ACC	gold-only	PPI	ACC	gold-only	PPI
HoC (bias −4.7 pts)							
50	0.000	0.966	0.968	0.936	0.652	0.239	**0.201**
100	0.000	0.950	0.950	0.922	0.463	0.169	**0.143**
200	0.000	0.942	0.960	0.964	0.284	0.120	**0.102**
300	0.000	0.944	0.968	0.960	0.217	0.098	**0.084**
ADE (bias −0.7 pts)							
50	1.000	0.944	0.888	0.912	0.449	0.199	**0.155**
100	1.000	0.944	0.912	0.930	0.307	0.141	**0.111**
200	0.994	0.952	0.954	0.934	0.181	0.101	**0.080**
300	0.994	0.968	0.960	0.964	0.144	0.083	**0.066**
HoC, zero-shot LLM predictor (bias −7.2 pts)							
50	0.000	0.956	0.958	0.948	0.780	0.238	**0.223**
100	0.000	0.956	0.946	0.938	0.696	0.169	**0.160**
300	0.000	0.918	0.950	0.964	0.451	0.098	**0.093**
ADE, zero-shot LLM predictor (bias +2.9 pts)							
50	0.000	0.928	0.900	0.908	0.407	0.200	**0.152**
100	0.000	0.922	0.928	0.914	0.249	0.142	**0.108**
300	0.000	0.922	0.952	0.956	0.120	0.082	**0.064**

D. Measured coverage on clinical labels

To evaluate coverage on clinical text, we treat the 123 physician labels as a finite population ($\theta^\star$=0.423), draw a representative gold subset of size $n_g \in \{30, 50, 70, 90\}$, and over 1,000 draws measure empirical 95% coverage. Table III reports the result for every zero-shot labeler at n_g=50 and Fig. 5(d) across n_g. The labelers' biases span -8.1 to +24.4 points on this population, which provides a controlled test of both error regimes. Naive counting uses no labels, so its interval is fixed and covers exactly when the bias is below the binomial half-width (8.7 points at N=123): it covers for the 6 labelers with bias at most 8.1 points and fails outright for the 2 with bias at least 13.8 points. PPI attains 0.934–0.953 coverage for every labeler at n_g=50 (0.92–0.96 across n_g) and gold-only 0.94–0.95; PPI is tighter than gold-only for every labeler and every n_g, by 12% on average. Under this representative finite-population design, PPI remains near

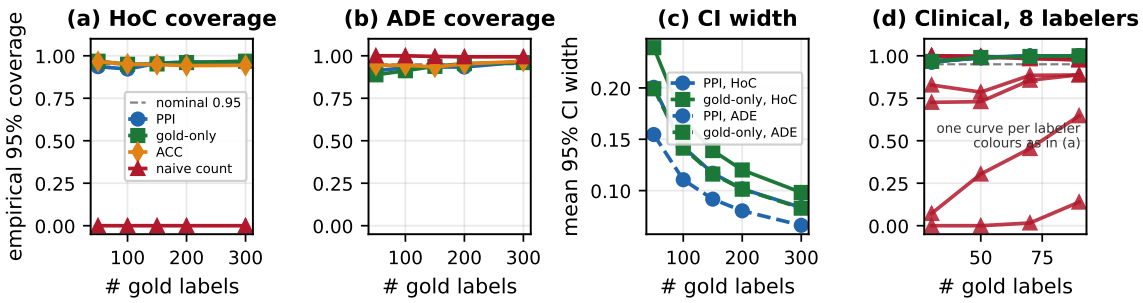


Fig. 5. Coverage validation. (a,b) On HoC and ADE (truth known) PPI intervals (blue) attain near-nominal 0.95 coverage of the true corpus rate, as do gold-only (green) and the ACC bootstrap interval (orange); naive counting (red) is at 0 on HoC, where the classifier is biased, and covers on ADE only because the classifier happens to be unbiased. (c) PPI intervals are narrower than gold-only at every n_g, by 16% (HoC) and 21% (ADE) on average. (d) Measured coverage on the clinical labels for every zero-shot labeler: treating the 123 physician labels as a finite population, re-drawing the gold set representatively, and applying the finite-population correction, PPI (blue, one curve per labeler) is near nominal for every labeler and gold-only (green) is valid, while naive counting (red) covers only for the labelers that happen to be nearly unbiased on this population (Sec. V-D).

TABLE III
MEASURED 95% COVERAGE AND MEAN WIDTH ON THE CLINICAL LABELS (n_g=50, 1,000 DRAWS). PPI IS NEAR NOMINAL FOR ALL EIGHT LABELERS AND TIGHTER THAN GOLD-ONLY THROUGHOUT.

		Coverage			Width	
Labeler	bias	naive	gold-only	PPI	gold-only	PPI
Qwen3.8-27B	−7.3	1.000	0.951	0.947	0.212	**0.195**
Qwen3-14B	+1.6	1.000	0.951	0.934	0.212	**0.181**
Qwen3-14B 4-bit	−0.8	1.000	0.951	0.939	0.212	**0.188**
Llama-3.1-8B	+2.4	1.000	0.951	0.939	0.212	**0.187**
Qwen3-30B-A3B	−8.1	1.000	0.951	0.939	0.212	**0.190**
Qwen3-32B	+13.8	0.000	0.951	0.945	0.212	**0.179**
Qwen2.5-72B (AWQ)	−1.6	1.000	0.951	0.953	0.212	**0.193**
Qwen3-14B†	+24.4	0.000	0.951	0.940	0.212	**0.178**

nominal across all evaluated labeler error profiles.

E. Ablation: robustness of the coverage result

On HoC (n_g=150, 500 draws), PPI coverage remains near nominal across the evaluated pseudo-label, power-tuning, calibration, and screening configurations (Table IV). Whether the pseudo-labels are fed as hard $\{0, 1\}$ decisions or Platt-calibrated probabilities, coverage stays near nominal (0.95–0.97) and width changes by under one point, with calibration giving the tightest interval. Across the power-tuning parameter, from λ=0 (gold-only) through λ=1 to the variance-minimizing `auto` (PPI++), coverage remains 0.93–0.97, and `auto` matches the tightest width. Finally, we apply the screen to HoC as on Long COVID: a recall-first threshold on the classifier score, chosen on the disjoint held-out split to keep 99% of positives, keeps 3,934 of 5,345 sentences (retaining 98.2% of positives) and changes the estimand to the screened pool's rate (0.339). Coverage of that estimand stays near nominal with the screen (0.94–0.95), as it does without it (0.97), with or without calibration.

When the assumption is violated: The guarantee can break in two ways, and we probe both on HoC. *Predictor fit on the labels.* If the predictor is trained or calibrated on the ground-truth labels used for estimation, its in-sample predictions are over-optimistic and naive PPI under-covers (0.86, width 0.099); cross-prediction-powered inference [3] with K=5 out-of-fold predictions restores nominal coverage (0.95) at width 0.120. This is the regime of our clinical deployment, which is why we cross-fit there. *Non-representative ground truth.* Oversampling positives into the gold set, so its positive rate climbs from the true 0.254 to 0.35 and 0.45, drops PPI coverage from 1.00 (stratified draws with a fixed positive count) to 0.52 and then to 0 (point estimate 0.25→0.32→0.39); gold-only coverage collapses faster still. Representativeness is necessary for either estimator.

TABLE IV
ROBUSTNESS OF COVERAGE ON HOC (n_g=150, 500 DRAWS).

Variant	Coverage	Width
pseudo-label: hard $\{0, 1\}$	0.946	0.123
pseudo-label: calibrated prob	0.966	0.117
λ=0 (gold-only)	0.934	0.139
λ=1	0.972	0.118
λ=`auto` (PPI++)	0.966	0.117
screen on, calibration off	0.948	0.136
screen on, calibration on	0.944	0.134

F. Choosing the retrieval operating point

Fig. 3 reports the precision/recall trade-off and the single fixed operating point. The screen is a union on the raw pool: a document is kept if the embedding similarity clears τ or any regex cue matches (Alg. 1), so the regex cues (alone recovering only 65% of gold positives at 48% precision) can only add candidates. At τ=0.40 the embedding keeps 2,281 of the 3,066 pool documents, the regex cues 679, and the union 2,383, pruning 22.3% of the raw pool while retaining every gold positive. Two caveats apply. First, τ is selected on $\mathcal{L}$ and its recall measured on the same 52 positives, so "retains every gold positive" holds only by construction on $\mathcal{L}$; we therefore report the pruning fraction as the substantive screen result. Second, because we score the full pool for every reported estimate, the screen enters no coverage or prevalence number; it is a deployment-time lever, and Sec. V-E shows that coverage survives it. The Qwen3-Reranker then sharpens the

TABLE V
TREATMENT AND ADVERSE-OUTCOME ASSOCIATIONS (551 LONG COVID CASE REPORTS). NO PAIR SURVIVES BENJAMINI–HOCHBERG CONTROL AT q=0.10 UNDER THE LEAKAGE-CONTROLLED BASELINE STRATIFICATION.

Treatment	n_t	lift	sev. lift	sev. rate [95% CI] (n)	non-sev. rate [95% CI] (n)
warfarin	8	2.18	1.04	1.00 [0.51,1.00] (4)	0.75 [0.30,0.95] (4)
piperacillin/tazobactam	10	1.99	1.25	0.83 [0.44,0.97] (6)	0.75 [0.30,0.95] (4)
broad-spectrum antibiotics	14	1.96	1.34	0.67 [0.35,0.88] (9)	1.00 [0.57,1.00] (5)
LMW heparin	11	1.81	1.14	1.00 [0.61,1.00] (6)	0.40 [0.12,0.77] (5)
atorvastatin	11	1.81	1.14	0.67 [0.30,0.90] (6)	0.80 [0.38,0.96] (5)
mechanical ventilation	31	1.77	1.08	0.75 [0.51,0.90] (16)	0.67 [0.42,0.85] (15)
vancomycin	24	1.77	1.04	0.67 [0.39,0.86] (12)	0.75 [0.47,0.91] (12)
meropenem	17	1.76	1.23	0.70 [0.40,0.89] (10)	0.71 [0.36,0.92] (7)

ranking of the retained documents for labeling order (AUPRC $0.655 \rightarrow 0.679$; at fixed 80% precision, recall rises $0.04 \rightarrow 0.27$).

G. Treatment–outcome associations: where the method stops

The rectifier does not transfer: PPI's rectifier (18) averages $y_j - f(x_j)$ over gold-labeled samples, so it corrects a statistic only when y labels the variable the statistic is about. Our 123 labels record Long COVID status; the association rates are functions of the extracted T (treatment) and O (adverse outcome), which no expert has labeled here. A coverage-valid version needs a representative expert-labeled sample of the extraction itself, from which PPI would estimate $P(T{=}1)$, $P(O{=}1)$ and $P(T{=}1, O{=}1)$, with delta-method or influence-function intervals for the derived conditional rates, differences and lift. We leave this extension to future work.

A stratification built from outcomes is circular: From 551 case reports we mine report-level co-occurrences for the 45 treatments mentioned in at least 8 reports, scoring each pair by *lift*, $P(t, o)/[P(t)P(o)]$ (base adverse rate 0.401), and splitting reports into severity strata. The natural severity proxy, and the one the accepted version used, reads severity off the outcome text (ICU, ventilation, respiratory failure, death). It leaks: 12 of those 14 markers also appear in the adverse-outcome lexicon, so for the 22% of reports whose severity is signaled by the outcome text $P(\text{adverse})$ is 1.00 by construction. Controlling the leakage—defining severity from pre-treatment baseline characteristics only, and not counting an outcome string that merely restates the tested treatment—collapses the apparent signal: no pair survives Benjamini–Hochberg control [44] at q=0.10 (adjusted q=1.00 for all 45 pairs), against 5 under the leaky outcome-derived stratification (Table V).

VI. CONCLUSION

We presented a label-efficient discovery methodology that turns a large noisy medical corpus and ~ 100 ground-truth labels into corpus-level statistics with valid confidence intervals, by feeding calibrated, cross-fitted LLM pseudo-labels into a prediction-powered estimator that is asymptotically never less efficient than expert-only inference. It delivers measured near-nominal coverage on two public benchmarks, where naive LLM-counting either fails entirely or succeeds only by luck and quantification baselines are several times wider, and transfers under a clearly stated representativeness assumption to a real biomedical corpus. Applied to treatment–outcome mining, after leakage control, no treatment–outcome pair survives FDR correction. So we report no confirmed pairs and leave coverage-valid association mining, which requires expert labels on the extractions themselves, to future work.

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